

Homework 07

REAL VARIABLES I

Fall 2026

Due: 2026-10-11 19:00

Topics:

Instructions

Submission is through Gradescope, accessible through [Blackboard](#).

Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%:** No submission or no meaningful content.
- **30%:** Attempt at solution, outline of idea.
- **60%:** Serious attempt at solution. Solution fails to address important points.
- **80%:** Solutions with fixable mistakes or incompleteness.
- **100%:** Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.

Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.

You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.

Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

1. (Rescaling and the Dirac delta)

1.a (weight: 1.0)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous bounded function on $\mathbb{R}$. Prove that

$$\frac{1}{\pi} \lim_{n \rightarrow \infty} \int_{\mathbb{R}} f(x) \frac{n}{1 + n^2 x^2} d\mathcal{L}(x) = C f(0)$$

where $C = \int_{\mathbb{R}} \frac{1}{1+x^2} d\mathcal{L}(x)$.

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Change variables $y = nx$.
- $\int_{-\infty}^{+\infty} \frac{1}{1+y^2} dy = \pi$.
- Do not forget to justify any limiting procedure.

2. (Lebesgue dominated convergence theorem)

2.a (weight: 0.3)

$$\lim_n \int_0^\pi \frac{\sin(nx)}{nx} dx$$

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Do not forget to justify any limiting procedure.

2.b (weight: 0.3)

$$\lim_n \int_0^\pi \frac{\sin(nx)}{nx} dx$$

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Do not forget to justify any limiting procedure.

2.c (weight: 0.7) **IN CLASS** Find

$$\lim_{n \rightarrow \infty} \int_0^n \left(1 - \frac{x}{n}\right)^n dx.$$

Free hint.

- You may assume that the Riemann and Lebesgue integrals coincide on continuous functions.
- Do not forget to justify any limiting procedure.

2.d (weight: 1)

3. (Integrals over subsets)

Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f_n : X \rightarrow [0, +\infty]$ be measurable and pointwise convergent to a function f . Assume

$$\int_X f_n(x) dx \rightarrow \int_X f(x) dx < \infty.$$

Prove that for any $E \subset X$

$$\int_E f_n(x) dx \rightarrow \int_E f(x) dx < \infty.$$

4. (Equiuniform integrability and Vitali's theorem)

Let $(X, \mathcal{A}, \mu)$ be a finite measure space, and let $f_n : X \rightarrow \mathbb{R}$ be a sequence of measurable functions that converges pointwise to a function f .

4.a (weight: 1) **IN CLASS**

Assume that

$$\sup_{n \in \mathbb{N}} \int_X |f_n(x)| \log(1 + |f_n(x)|) d\mu(x) < +\infty$$

. Prove that

$$\lim_n \int_X f_n(x) d\mu(x) = \lim_n \int_X f(x) d\mu(x)$$

Free hint. Prove that the sequence f_n is equi-uniformly integrable. Apply Vitali's convergence theorem.